TECHNICAL ARCHITECTURE & INDUSTRY SOLUTION BRIEF

Rouen for the Energy & Utilities Industry

Bridging isolated OT field assets, legacy industrial protocols, and central control rooms using Outbound-Only WebSocket Mesh Federation and Zero-Compile Adaptive Process Cards.

ISA/IEC 62443 & NERC CIP Friendly | Outbound-Only WSS (Port 443) | Ed25519 Mutual Auth | Universal stdio Sensor Bridge | Native C++20 Footprint

01 The Energy OT/IT Challenge

Modern power grids, utility substations, Distributed Energy Resources (DERs), Battery Energy Storage Systems (BESS), wind/solar farms, and pipeline telemetry networks face two structural bottlenecks when scaling real-time visibility and operator control:

1. Strict Network Isolation (The OT/IT Boundary)

- **No Inbound Connections:** Substations and generation sites enforce unidirectional firewall rules under NERC CIP and ISA/IEC 62443. Opening inbound ports to field networks is prohibited.
- **CGNAT Cellular & Satellite Links:** Remote solar sites, reclosers, and wellheads rely on 4G/5G cellular modems or Starlink behind Carrier-Grade NAT with no routable public IP address.
- **VPN Fragility:** Managing hundreds of site-to-site IPsec/WireGuard tunnels across multi-vendor customer sites creates high operational overhead and lateral-movement risk.

2. Legacy Protocol & Hardware Fragmentation

- **Multi-Decade Equipment Mix:** Field assets speak Modbus RTU/TCP, DNP3, IEC 61850, OPC-UA, BACnet, CAN bus, and proprietary RS-485 serial commands.
- **Heavyweight SCADA Lock-In:** Traditional HMI/SCADA platforms require proprietary driver licenses, compiled C++/C# plugins, or heavy JVM/container stacks unsuitable for DIN-rail edge PCs.
- **Slow Field Iteration:** Swapping an inverter or protection relay model shouldn't require recompiling or redeploying the entire operator console.

02 Pillar 1: Outbound-Only WebSocket Mesh (`rouen_mesh_host`)

Implemented in `rouen::mesh::rouen_mesh_host`, every field node connects to the Rouen Mesh via a single outbound TLS WebSocket (`wss://` on port 443). Field gateways never open listening ports on untrusted networks. Identity is cryptographically enforced via mutual Ed25519 challenge-response signatures, multiplexing card proxying (`HTTP_REQUEST` / `HTTP_RESPONSE`), live ephemeral state registries (`REGISTRY_SET`), and auditable point-to-point streams (`ROUTE_OPEN`).

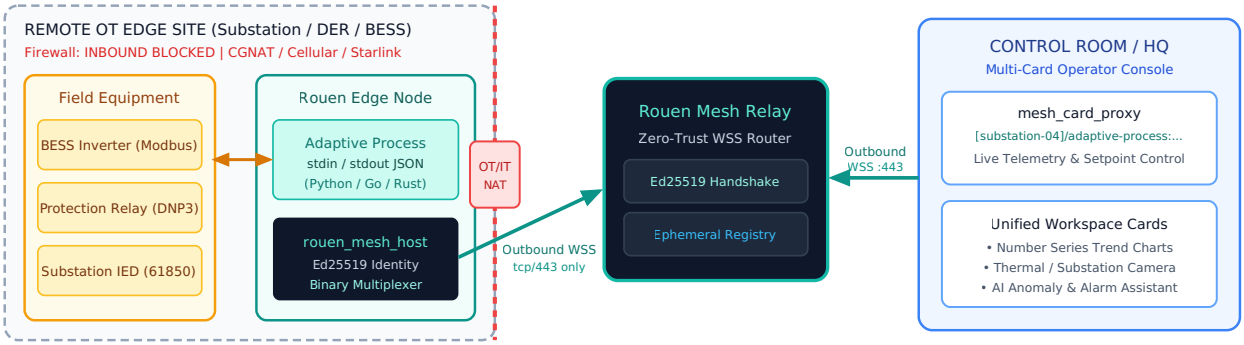


Figure 1 — Zero-inbound-port OT/IT topology: Both the remote substation node and the operator console establish outbound-only TLS WebSockets. Raw Modbus/DNP3/IEC 61850 traffic never leaves the substation LAN.

03 Pillar 2: Universal Sensor Integration (`adaptive_process_card`)

Implemented in `rouen::cards::adaptive_process_card` on top of `rouen::helpers::piped_process`, the `adaptive-process:<command>` URI scheme turns any executable or script in any language into a live, two-way interactive HMI card without compiling against Rouen or matching a C++ ABI:

- `stdout` Stream (Telemetry Emission): Each newline-delimited JSON line emitted by the child process renders immediately as an Adaptive Card UI, supporting automatic cadence polling via `"refreshIntervalMs"`.
- `stdin` Stream (Operator Commands): Clicking `Action.Submit` or `Action.Execute` on the card serializes all form inputs and action verbs as a single JSON line written directly to the process's `stdin`.
- Crash Isolation & Live `stderr` Capture: Faulty serial drivers or vendor SDK exceptions are isolated in the child OS process, with live diagnostic output captured in a collapsible Process stderr drawer.

04 The "Killer Combo": Mesh Card Proxy + Adaptive Process

SEAMLESS EDGE-TO-CONTROL-ROOM COMPOSITION

By combining `mesh_card_proxy` with `adaptive_process_card`, an operator at HQ opens a live interactive card backed by a hardware script executing directly on a remote substation or DER gateway:

```
[substation-west-04]/adaptive-process:python3 /opt/grid/modbus_bess_controller.py --unit 1
```

Why it wins in Energy: The Python/Go driver executes right next to the PLC/inverter with sub-millisecond local bus latency, while the operator interacts with a responsive, cryptographically authenticated UI card across the mesh—with zero cloud vendor lock-in.

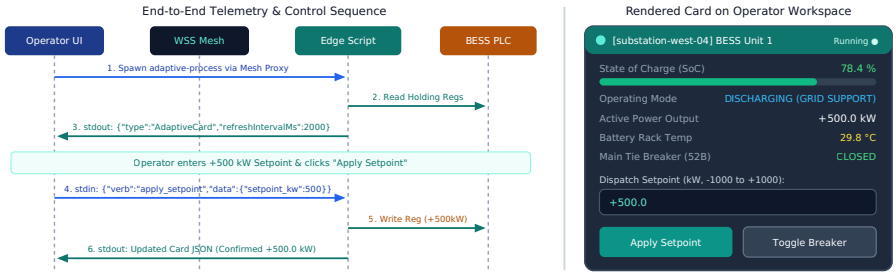


Figure 2 — Left: Bidirectional stdio-over-Mesh protocol flow. Right: The resulting live Adaptive Process Card rendered in Rouen.

05 Reference Implementation: Python BESS Edge Bridge

A complete, dependency-free edge script requires ~35 lines of Python. Any language capable of reading `stdin` and printing newline-delimited JSON to `stdout` works identically:

```python
#!/usr/bin/env python3  —  Launch remotely: [substation-west-04]/adaptive-process:python3 /opt/rouen/bess_card.py
import json, sys, time
state = {"site": "Substation West 04 - BESS Unit 1", "soc": 78.4, "kw": -250.0, "set_kw": -250.0, "closed": True}

def emit_card(note="Nominal"):
    mode = "DISCHARGING" if state["kw"] > 0 else ("CHARGING" if state["kw"] < 0 else "IDLE")
    card = {"type": "AdaptiveCard", "version": "1.5", "refreshIntervalMs": 2000, "body": [
        {"type": "TextBlock", "text": state["site"], "size": "large", "weight": "bolder"},
        {"type": "FactSet", "facts": [
            {"title": "State of Charge", "value": f"{state['soc']:.1f} %"},
            {"title": "Active Power",    "value": f"{state['kw']:+.1f} kW ({mode})"},
            {"title": "Main Breaker",    "value": "CLOSED" if state["closed"] else "OPEN"},
            {"title": "Status",          "value": f"{time.strftime('%H:%M:%S')} ({note})"}]},
        {"type": "Input.Number", "id": "set_kw", "title": "Dispatch Setpoint (kW)", "value": state["set_kw"]}
    ], "actions": [
        {"type": "Action.Execute", "title": "Apply Setpoint", "verb": "apply_setpoint"},
        {"type": "Action.Execute", "title": "Toggle Breaker", "verb": "toggle_breaker"}]}
    print(json.dumps(card), flush=True)

emit_card("Initial poll")
for raw in sys.stdin:
    msg = json.loads(raw.strip() or "{}"); verb, data = msg.get("verb", ""), msg.get("data", msg)
    if verb == "apply_setpoint": state["set_kw"] = float(data.get("set_kw", state["set_kw"]))
    elif verb == "toggle_breaker": state["closed"] = not state["closed"]
    state["kw"] = state["set_kw"] if state["closed"] else 0.0; emit_card(verb or "Live poll")
```

06 High-Value Energy & Utilities Use Cases

ENERGY SUB-SECTOR	OPERATIONAL PAIN POINT	ROUEN MESH + ADAPTIVE PROCESS ADVANTAGE
DER & Microgrid Fleet Operations	Hundreds of behind-the-meter commercial solar + BESS sites behind customer firewalls and cellular CGNAT.	Outbound-only `wss://` links every site to a single operator workspace with zero VPN appliances or inbound port rules.
Substation & Protection Commissioning	Field engineers run custom Python/IEC 61850/DNP3 scripts on laptops while coordinating with senior engineers at HQ.	Wrap existing field scripts as `adaptive-process:` cards; HQ engineers view and co-operate the live test card over the mesh.
Upstream & Pipeline Telemetry	High-latency, metered satellite/cellular links where bloated web dashboards fail or consume excessive data quotas.	Compact single-line JSON updates over persistent binary WebSocket frames minimize bandwidth while keeping interactive control.
Grid Lab & HIL Testbenches	Combining thermal cameras, oscilloscopes, power analyzers, and custom serial rigs across multiple workstations.	Unifies Rouen's native Camera, Number Series, and Process Orchestrator cards alongside Adaptive Process instrument drivers.